



ANALYSTLAB AFRICA

Experience Lab Internship Programme

Official Weekly Assignment

Week 6: HealthConnect Integration, Advanced Development & Validation

Document Control

Programme	AnalystLab Africa Experience Lab
Week	Week 6
Project	HealthConnect Clinic Experience Lab
Project Title	Improving Patient Appointment Attendance and Healthcare Support Using Data and AI
Version	1.0
Issue Date	_____
Submission Deadline	End of Week 6 (Sunday, 11:59 PM WAT)
Submission Method	Official AnalystLab Africa Week 6 Submission Form

1. BUSINESS SCENARIO

HealthConnect Clinic is a fictional healthcare provider seeking to reduce missed appointments, improve appointment attendance, make better use of appointment slots and provide more effective administrative support to patients.

During **Week 4**, interns established their understanding of the HealthConnect problem, reviewed the available resources and defined their track-specific responsibilities.

During **Week 5**, interns moved from planning into practical implementation and produced initial project outputs across their respective professional tracks.

These outputs now provide the foundation for the next stage of the project. Week 5 specifically required interns to identify meaningful cross-track dependencies and prepare their work for the next stage.

Week 6 moves the project into:

Integration → Advanced Development → Validation

The focus is no longer simply on producing an individual component.

Interns must now demonstrate **how their work connects with the broader HealthConnect solution**, improve the quality of their outputs and validate whether those outputs are suitable for the next stage of the project.

The overall project question remains:

How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?

2. PROJECT CONTINUITY

From Week 4 onwards, all interns contribute to the same HealthConnect Experience Lab while working from the perspective of their respective professional tracks.

The project now progresses as follows:

Week 4

Problem Understanding → Resource Review → Solution Planning

Week 5

Analysis → Development → Initial Implementation

Week 6

Integration → Advanced Development → Validation

Week 7

Testing → Refinement → End-to-End Validation

Week 8

Final Integration → Presentation

Important

You must **continue from your Week 5 work**.

Do not:

- Start a new HealthConnect project.
- Select a new dataset.
- Discard your Week 5 work.
- Repeat your complete Week 5 analysis.
- Rebuild an identical Week 5 deliverable without meaningful improvement.

Your Week 6 work should show **progress from what you produced in Week 5**.

3. WEEK 5 → WEEK 6 TRANSITION

Before beginning the new work, briefly establish the connection between the two weeks.

Identify:

1. Your main Week 5 output.
2. The most important result/component produced.
3. The main issue or limitation discovered.
4. The track(s) whose work is relevant to your Week 6 activities.
5. What you intend to improve, integrate or validate in Week 6.

Do not reproduce your Week 5 report.

This section should be a concise transition into your Week 6 work.

4. WEEK 6 ASSIGNMENT OBJECTIVE

The objective of Week 6 is to **advance and connect the components developed during Week 5 while validating their quality, usability and relevance to the overall HealthConnect project.**

By the end of Week 6, interns should be able to:

- Improve an existing project component.
- Work with outputs from another track where relevant.
- Demonstrate meaningful cross-track integration.
- Validate their work against defined requirements.
- Identify and resolve integration issues.
- Document technical, analytical or project decisions.
- Evaluate the limitations of their current solution.
- Produce evidence that their work is ready for the next testing stage.

5. WEEK 6 LEARNING OBJECTIVES

By completing this assignment, interns should be able to:

- Integrate work within a multidisciplinary project.
- Apply feedback and findings from previous project activities.
- Improve existing technical or analytical solutions.
- Validate outputs using appropriate methods.
- Identify errors, inconsistencies and integration issues.
- Communicate dependencies professionally.
- Document changes and decisions.
- Evaluate whether a solution component meets its intended purpose.
- Contribute meaningfully to a shared project.

6. PROJECT RESOURCES

Continue using the approved HealthConnect project resources:

6.1 HealthConnect Appointment Dataset

HealthConnect_Appointment_Data.csv

6.2 HealthConnect Data Dictionary

HealthConnect_Data_Dictionary.xlsx

6.3 HealthConnect Clinic Knowledge Base

HealthConnect_Clinic_Knowledge_Base.docx

The Week 5 document establishes these as the continuing project resources.

Important

Do not overwrite the original project resources.

Any processed, transformed or derived files must remain separate and properly documented.

For Generative AI work, responses must remain within the approved HealthConnect Knowledge Base and must not invent clinic or medical information.

7. PART 1 — WEEK 6 INTEGRATION READINESS

Before performing your track-specific work, identify the **specific project connection** you will work on during Week 6.

Complete the following:

A. Identify your integration point

- Which track are you working with?
- What output do you need from that track?
- What output will you provide to that track?
- Why is the connection relevant?

B. Define the integration objective

Clearly state:

What should be different or improved after this integration activity?

C. Establish evidence

Your collaboration must produce evidence such as:

- Updated file
- Analysis output
- Model artefact
- Code
- Dashboard update
- Prototype update
- Project documentation
- Testing result
- Meeting/coordination record
- GitHub commit where applicable

Simply communicating with another intern does not constitute integration.

8. PART 2 — TRACK-SPECIFIC ASSIGNMENTS

A. PROJECT MANAGEMENT TRACK

Week 6: Project Integration & Coordination

Week 5 focused on project execution, progress tracking, risks, dependencies and coordination.

Week 6 moves the Project Management role toward **coordinating integration between the technical workstreams**.

Tasks

1. Create a **HealthConnect Workstream Integration Map** showing how the different tracks connect.
2. Identify the major dependencies between project workstreams.
3. Identify which deliverables are ready for integration.
4. Identify deliverables or activities that require further development.
5. Track cross-track integration activities.
6. Establish an integration issue log.
7. Record major project decisions made during Week 6.
8. Monitor whether project activities remain aligned with the HealthConnect objective.
9. Update the project timeline based on actual progress.
10. Review and update project risks and dependencies based on integration activities.
11. Track unresolved issues requiring attention from specific tracks.
12. Coordinate communication between relevant tracks.
13. Prepare a Week 7 testing and refinement coordination plan.

Week 6 Output

HealthConnect Project Integration & Coordination Package

Include:

- Workstream Integration Map
- Updated project timeline
- Integration/dependency register
- Integration issue log
- Decision log
- Updated risk register
- Cross-track coordination record
- Project status update
- Week 7 testing coordination plan

B. DATA ANALYTICS TRACK

Week 6: Advanced Analytics & Decision Support

Week 5 already required data preparation, EDA, KPI calculations, dashboard/visualisation and at least five business insights.

Do not repeat the complete Week 5 EDA or simply recreate the same dashboard.

Week 6 should focus on **deepening and validating the findings produced in Week 5 and using them to support project decisions.**

Tasks

1. Select the most important findings from Week 5 that require deeper investigation.
2. Conduct additional analysis to determine whether these findings remain meaningful.
3. Investigate important patterns across relevant patient, appointment or operational segments.
4. Analyse relationships that could help explain appointment attendance or no-show behaviour.
5. Validate the most important KPIs developed in Week 5.
6. Identify which findings have the greatest potential business impact.
7. Translate significant findings into specific HealthConnect actions.
8. Provide relevant analytical findings to the Data Science track where they can support modelling decisions.
9. Review whether any Week 5 conclusions need to be revised based on the deeper analysis.
10. Improve the existing dashboard or analytical report based on the new findings.
11. Document the evidence supporting your key recommendations.
12. Document your cross-track analytical contribution.

Week 6 Output

HealthConnect Advanced Analytics & Decision Support Report/Notebook

Include:

- Advanced analysis
- Validated findings
- Refined KPIs
- Updated visualisations/dashboard
- Evidence-based insights
- Business recommendations

- Data limitations
- Cross-track contribution
- Week 7 analytical testing requirements

C. DATA SCIENCE TRACK

Week 6: Model Improvement, Error Analysis & Validation

Week 5 established the **baseline classification model** and initial evaluation. The Week 5 document explicitly states that the objective at that stage was to establish a reliable baseline rather than achieve the final best-performing model.

Therefore, **do not simply train the same baseline model again.**

Tasks

1. Analyse the weaknesses of your Week 5 baseline model.
2. Identify the most important sources of model error.
3. Conduct error analysis using appropriate evaluation techniques.
4. Investigate false positives and false negatives where applicable.
5. Review whether the current features adequately support the prediction objective.
6. Refine or replace features where there is a justified reason.
7. Develop at least one improved modelling approach.
8. Compare the improved approach with the Week 5 baseline.
9. Evaluate the competing model(s) using appropriate metrics.
10. Assess whether the model performance is suitable for the intended HealthConnect use case.
11. Incorporate relevant findings from the Data Analytics track where appropriate.
12. Document the final candidate model for the next stage.
13. Clearly explain any modelling decisions made during Week 6.
14. Document model limitations and risks.
15. Define the model testing requirements for Week 7.

Week 6 Output

HealthConnect Model Improvement & Validation Notebook/Report

Include:

- Baseline performance summary
- Error analysis
- Feature refinement
- Improved model(s)
- Model comparison
- Evaluation results
- Interpretation
- Business relevance
- Candidate model recommendation
- Limitations
- Cross-track contribution
- Week 7 testing requirements

D. MACHINE LEARNING ENGINEERING TRACK

Week 6: ML Pipeline Integration & Validation

Week 5 established the initial data-processing workflow, reusable preprocessing components, model workflow/integration structure, configuration requirements, documentation and basic component testing.

Week 6 should therefore focus on **integrating and validating the components rather than rebuilding the Week 5 pipeline.**

Tasks

1. Review the current Week 5 pipeline structure and identify integration gaps.
2. Integrate the appropriate Data Science model or model interface into the pipeline.
3. Ensure that data processing and model input/output are compatible.
4. Validate the flow of data through the integrated pipeline.
5. Implement appropriate validation checks for pipeline inputs and outputs.
6. Test the interaction between preprocessing and model components.
7. Identify and resolve integration errors.
8. Improve pipeline reliability where weaknesses are identified.
9. Improve configuration or environment handling where required.
10. Add appropriate logging or execution information.
11. Verify that the pipeline can be reproduced using the documented setup.
12. Update the repository structure and documentation where necessary.
13. Document technical dependencies on other tracks.
14. Prepare the integrated pipeline for broader testing in Week 7.

Week 6 Output

HealthConnect Integrated ML Pipeline Package

Include:

- Integrated pipeline
- Model integration
- Data/model interface documentation
- Validation checks
- Integration testing evidence
- Updated repository
- Configuration/dependency documentation
- Issue log
- Updated technical documentation
- Week 7 testing plan

Important: Week 6 is still not the final production deployment stage.

E. GENERATIVE AI TRACK

Week 6: AI Assistant Evaluation, Refinement & Reliability

Week 5 already required the creation of the initial assistant prototype, prompt library, test cases, Knowledge Base usage approach, safety testing and escalation testing.

Therefore, Week 6 should **not simply create another prototype**.

The focus is now:

Evaluate → Identify weaknesses → Improve → Validate

Tasks

1. Analyse the results of the Week 5 assistant tests.
2. Identify responses that were inaccurate, incomplete, unclear or inconsistent.
3. Identify weaknesses in the existing prompt structure.
4. Refine prompts based on observed weaknesses.
5. Develop additional test scenarios based on realistic HealthConnect interactions.
6. Evaluate the assistant's consistency across similar questions.
7. Test the assistant's ability to remain within the approved Knowledge Base.
8. Test situations where the Knowledge Base does not provide sufficient information.
9. Test escalation behaviour for requests requiring escalation.
10. Test unsupported requests and verify that the assistant responds appropriately.
11. Improve the assistant workflow based on evaluation results.
12. Document examples of improvements between the Week 5 and Week 6 versions.
13. Define measurable criteria for evaluating the assistant during Week 7.
14. Document remaining safety, reliability or knowledge limitations.
15. Document the assistant's cross-track integration point where applicable.

Week 6 Output

HealthConnect AI Assistant Evaluation & Refinement Package

Include:

- Refined prompt library
- Expanded test set
- Evaluation results
- Before/after examples
- Error/weak-response analysis
- Improved assistant workflow/prototype
- Knowledge Base compliance evidence
- Safety and escalation validation
- Limitations
- Cross-track contribution
- Week 7 testing plan

9. MANDATORY CROSS-TRACK INTEGRATION

This is the **major difference between Week 5 and Week 6.**

Week 5 asked interns to **identify** at least one meaningful dependency or collaboration point.

In Week 6, interns must actually use that connection.

Each intern must complete **at least one meaningful cross-track integration activity.**

Document:

1. Track collaborated with:

2. Project dependency:

3. Information/output received:

4. Information/output provided:

5. Integration activity completed:

6. What changed as a result:

7. Evidence:

Important

Communication alone is not sufficient.

The collaboration must result in a meaningful change, improvement, decision, integration or validated output.

10. HEALTHCONNECT INTEGRATION MATRIX

Use the following as guidance:

Track	Potential Integration
Data Analytics → Data Science	Analytical findings inform modelling decisions
Data Science → ML Engineering	Candidate model and model requirements inform pipeline integration
ML Engineering → Data Science	Technical integration requirements inform model implementation
Data Analytics → Project Management	Findings and analytical progress inform project decisions
GenAI → Project Management	Assistant progress, issues and dependencies
Project Management → All Tracks	Coordinates integration, dependencies, risks and timelines
All Tracks → All Tracks	Relevant information required to advance HealthConnect

These are **examples**, not instructions that every intern must follow exactly.

Your actual integration should be based on your Week 5 work.

11. INTEGRATION VALIDATION

By the end of Week 6, every intern must answer:

What did I have at the end of Week 5?

↓

What did I improve in Week 6?

↓

Which other track did I work with?

↓

What did I receive?

↓

What did I provide?



What changed because of the integration?



How does this improve the overall HealthConnect project?

This should be supported with evidence.

12. ASSUMPTIONS, LIMITATIONS, RISKS & DEPENDENCIES

Review the issues identified during Weeks 4 and 5.

Document:

- Issues that have been resolved.
- Issues that remain unresolved.
- New issues discovered.
- New cross-track dependencies.
- Technical limitations.
- Data limitations.
- Modelling limitations.
- AI safety/reliability concerns where applicable.
- Integration challenges.
- Potential impact on the project.
- Recommended mitigation.

Do not create a separate report if these items can be incorporated into your main Week 6 deliverable.

13. WEEK 6 PROJECT SUMMARY

Prepare a concise Week 6 Project Summary covering:

1. What you planned to accomplish.
2. What you completed.
3. What you improved from Week 5.
4. What you integrated.
5. Which track(s) you collaborated with.
6. What was exchanged.
7. What changed as a result.
8. Key findings or development outcomes.
9. Major challenges.
10. Important decisions.
11. Remaining issues.
12. Your contribution to the overall HealthConnect project.

13. Your proposed focus for Week 7.

The summary should **complement your main deliverable and should not repeat it.**

14. SUBMISSION REQUIREMENTS

Your Week 6 submission must contain:

1. Track-Specific Week 6 Output

Submit the required deliverable for your assigned track.

2. Week 6 Project Summary

Submit the concise project summary.

3. Cross-Track Integration Evidence

Provide evidence of your Week 6 collaboration/integration activity.

4. Supporting Files

Where applicable:

- Reports
- Notebooks
- Dashboards
- Code
- Models
- Pipelines
- Prototypes
- Testing evidence
- Documentation
- Project management artefacts

5. Google Drive

Upload the required files to your designated project folder.

6. GitHub

Where applicable:

- Commit your Week 6 work.
- Update README.md.
- Maintain an organised repository.
- Ensure relevant files are accessible.

15. PROFESSIONAL DEVELOPMENT REQUIREMENT

Continue maintaining your HealthConnect project professionally.

Interns should:

- Maintain organised project folders/repositories.
- Document important decisions.
- Maintain appropriate version control.
- Clearly document changes between weeks.
- Update README.md where applicable.
- Present their work professionally.
- Share meaningful project progress on LinkedIn/X where appropriate.
- Accurately represent their actual contribution.

Assessment Criteria

Assessment Criteria	Weight
Application of Previous Project Work	15%
Track-Specific Development & Quality	25%
Integration & Validation	20%
Problem-Solving & Decision-Making	15%
Cross-Track Collaboration	10%
Documentation & Organisation	10%
Professional Presentation	5%
Total	100%

17. IMPORTANT WEEK 6 REMINDER

WEEK 6 IS NOT A REPEAT OF WEEK 5.

You are expected to **build forward**, not start again.

Your progression should be:

Week 5 Output → Improve → Integrate → Validate → Document → Prepare for Week 7

You are **not expected to complete the entire HealthConnect solution this week.**

The purpose of Week 6 is to demonstrate that your individual contribution can begin to function as part of a **larger multidisciplinary HealthConnect project.**

Your work should show:

Individual Expertise + Cross-Track Integration + Evidence-Based Improvement

The key difference between Week 5 and Week 6

Week 5	Week 6
Initial EDA	Advanced/decision-support analysis
Calculate KPIs	Validate/refine important KPIs
Initial dashboard	Improve dashboard based on findings
Baseline model	Improve and compare models
Initial model evaluation	Error analysis & validation
Initial ML pipeline	Integrated pipeline
Initial AI prototype	AI evaluation & refinement
Identify collaboration	Actually integrate with another track
Track-specific development	Track-specific development + integration
Prepare for Week 6	Prepare for Week 7 testing